



VMware vSphere infrastructure Enterprise Virtualization Architecture

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Chapter 1

VMware vSphere

1.1 Introduction to VMware vSphere

VMware vSphere is a powerful cloud computing virtualization platform developed by VMware, serving as the backbone of modern data centers. It provides a robust framework by:

- Abstracting physical hardware resources into dynamic, software-defined entities.
- Decoupling computing, storage, and networking resources from the underlying hardware.
- Enhancing resource utilization, management efficiency, and flexibility.

At its core, vSphere comprises two key components:

- **VMware ESXi:** Functions as the hypervisor, creating and running Virtual Machines (VMs) on physical servers.
- **VMware vCenter:** Acts as the centralized management platform, allowing administrators to oversee multiple ESXi hosts within a network and integrating advanced vSphere features for seamless administration, automation, and scalability.

By managing infrastructures as a unified operating environment, vSphere empowers organizations to build highly efficient, scalable, and secure virtualized ecosystems, making it an essential component of modern cloud computing and enterprise IT environments.

1.2 vSphere Core Components and Features

VMware vSphere is a comprehensive suite of virtualization and management tools designed to transform data centers into efficient, flexible, and resilient infrastructures. Here are its core components and features:

1. **Hypervisor:** Central to vSphere is the hypervisor, a lightweight software layer that facilitates the virtualization of physical servers. VMware's ESXi hypervisor boasts unparalleled performance, reliability, and security, laying the foundation for virtualized environments.
2. **vCenter Server:** Acting as the nerve center of vSphere deployments, vCenter Server provides a centralized platform for managing and orchestrating virtualized infrastructure. From provisioning and monitoring to automation and policy enforcement, vCenter Server simplifies administrative tasks while offering granular control over resources.

3. **High Availability (HA):** Downtime is unacceptable in today's always-on world. vSphere HA ensures continuous uptime by automatically restarting virtual machines on alternate hosts in the event of hardware failures, minimizing disruptions and preserving service availability.
4. **Distributed Resource Scheduler (DRS):** Resource management becomes a breeze with DRS, which dynamically balances computing workloads across hosts within a cluster. By optimizing resource utilization and load distribution, DRS enhances performance, scalability, and resilience.
5. **Storage vMotion:** Like vMotion for compute, Storage vMotion enables live migration of VM disk files between storage devices without disrupting VM operation. This feature facilitates storage maintenance, load balancing, and storage upgrades with minimal impact on VM.
6. **vSphere Distributed Switch (vDS):** vDS is a centralized and distributed network switch that simplifies network configuration and management across multiple hosts. It provides advanced networking features such as traffic shaping, load balancing, and Virtual Local Access Network (VLAN) tagging, enhancing network performance and scalability.
7. **Fault Tolerance (FT):** FT ensures continuous availability of VMs by maintaining a synchronized secondary VM that mirrors the primary VM's execution. In the event of a host failure, the secondary VM seamlessly takes over without interruption, providing continuous protection for critical workloads.
8. **vSphere Replication:** vSphere Replication enables asynchronous replication of VMs between hosts and datastores for disaster recovery purposes. It provides flexible and cost-effective replication options for both onsite and offsite data protection.
9. **Virtual Machine Templates and Cloning:** vSphere enables the creation of VM templates, which serve as master copies for deploying new VM instances. Templates streamline the VM provisioning process by preconfiguring settings such as OS, applications, and networking. Additionally, vSphere supports VM cloning, allowing administrators to create copies of existing VMs for rapid
10. **vSphere Networking Features:** In addition to the vSphere Distributed Switch (vDS), vSphere offers advanced networking features such as Network I/O Control (NIOC) for prioritizing network traffic, and VMware NSX integration for software-defined networking (SDN) capabilities such as micro-segmentation and network virtualization.
11. **vSphere Virtual SAN (vSAN):** vSAN is a software-defined storage solution integrated into vSphere, which aggregates local storage devices from ESXi hosts to create a shared storage pool. vSAN provides distributed, resilient storage with features like data deduplication, compression, and erasure coding, enabling organizations to leverage cost-effective storage solutions while ensuring high performance and availability for virtualized workloads.
12. **vSphere Lifecycle Management:** vSphere simplifies lifecycle management tasks such as hardware provisioning, firmware updates, and patch management through automation and centralized control. This includes features like vSphere Lifecycle Manager (vLCM), which provides a single interface for managing ESXi hosts, firmware, and driver updates.

13. **vSphere Content Library:** The vSphere Content Library simplifies VM and template management by centralizing content such as VM templates, ISO images, and scripts in a single, easily accessible repository. Content libraries can be shared across multiple vCenter Server instances, enabling consistent content management and deployment across distributed environments.

1.3 ESXi the Key to Virtualization

Core Functionality and Benefits: VMware ESXi is a high-performance Type 1 hypervisor that operates directly on physical servers, enabling the consolidation of multiple virtual machines onto shared hardware. As a central component of the VMware vSphere suite, it delivers a stable and efficient platform for managing virtualized environments, offering enhanced performance, improved security, and optimized resource utilization by eliminating the need for a traditional underlying operating system.

Resource Management and Operational Advantages: ESXi empowers organizations to create and manage virtual machines, each with its own dedicated CPU, memory, storage, and network resources. This efficient allocation maximizes hardware usage and contributes to significant cost reductions. Furthermore, ESXi enhances system reliability and operational flexibility through key features like vSphere High Availability (HA) for automatic recovery from server failures and vMotion for live migration of virtual machines without any service interruption, facilitating maintenance, load balancing, and robust disaster recovery capabilities.

Security and Scalability for Diverse Environments: Security is a paramount concern in ESXi, evidenced by integrated features such as secure boot to ensure system integrity, virtual machine encryption to protect sensitive data, and comprehensive access control mechanisms to safeguard against unauthorized access. Additionally, ESXi is highly scalable, capable of supporting large and complex infrastructures with numerous servers and substantial computing power, establishing it as a dependable virtualization solution for a wide array of deployments, including enterprise-level businesses, large-scale data centers, and dynamic cloud computing platforms.

1.3.1 What Are the Essential Files for a Virtual Machine (VM)?

When creating and managing virtual machines in VMware vSphere, several essential files are generated to store the VM's configuration, virtual hardware, and operational data. These files play a crucial role in the functionality and management of virtualized environments.

Core VM Files

- **VM Configuration File (.vmx):** Contains the fundamental settings and configurations of the VM, including hardware specifications, memory allocation, and network settings.
- **Virtual Disk File (.vmdk):** Acts as the virtual hard drive of the VM, storing the operating system, applications, and data.

Files Generated During VM Operation

- **VM Log File (vmware.log):** This file records system activities and events related to the VM's operation. If the log reaches its maximum size, a new log file is generated, and older logs are eventually removed.

- **NVRAM File (.nvram):** This file stores the BIOS/UEFI configuration of the VM, ensuring that firmware settings remain persistent across reboots.
- **Backup Configuration File (.vmx):** This file acts as a temporary backup of the primary VM configuration file.

1.4 VMware vCenter

VMware vCenter is a robust management platform designed to provide a centralized solution for overseeing virtualized infrastructures. It plays a crucial role in resource provisioning, performance monitoring, and operational management of both VMs and ESXi hosts. By consolidating these management functions into a single interface, vCenter enhances the efficiency and scalability of modern data centers.

1.4.1 Core Architecture and Functionality

vCenter is deployed as a VM and is built on Photon OS, a lightweight and optimized Linux-based operating system. This makes it highly efficient and secure. Within this VM, vCenter also includes the vCenter Server Application, which serves as the central software responsible for managing ESXi hosts and VMs. Additionally, it features an embedded database, which ensures stability and high performance when handling large-scale virtualization tasks.

1.4.2 Transition from Windows to Linux

VMware transitioned from Windows-based vCenter (up to vSphere 6.7) to Linux-based Photon OS in later versions, significantly reducing deployment time, operational costs, and improving stability and security.

1.4.3 Scalability and Performance

vCenter is designed for large-scale environments, capable of managing:

- Up to 2,500 ESXi hosts
- 45,000 registered VMs
- 40,000 powered-on VMs

This high scalability, combined with centralized management, optimizes resource allocation and enhances system resilience.

Size	Hosts - VMs	vCPUs	Memory	Storage
Tiny	Up to 10 Hosts or 100 VMs	2	12 GB	579 GB
Small	Up to 100 Hosts or 1000 VMs	4	19 GB	694 GB
Medium	Up to 400 Hosts or 4000 VMs	8	28 GB	908 GB
Large	Up to 1000 Hosts or 10000 VMs	16	37 GB	1358 GB
X-Large	Up to 2500 Hosts or 45000 VMs	24	56 GB	2283 GB

Figure 1.1: Sizing Guidelines Based on Environment Scale

1.4.4 Key Features of VMware vCenter

vSphere networking is a core component of VMware vSphere that enables communication between VMs, ESXi hosts, and external networks. It provides flexibility, security, and performance optimization for virtualized environments.

- **Core Management & Resource Optimization:**

- Provides a unified interface for managing multiple ESXi hosts, VMs, and clusters, eliminating manual host-by-host administration.
- Maintains a detailed inventory of resources.
- Optimizes resources using tools like DRS and Storage DRS, ensuring balanced workload distribution.

- **High Availability & Live Migration:**

- Enhances system resilience through HA and FT, minimizing downtime during failures.
- Enables live VM migrations between hosts without service interruption using vMotion, facilitating maintenance and disaster recovery without disruptions.

- **Automation & Scalability:**

- Supports automation via APIs and scripting tools, streamlining deployments and routine maintenance.
- Scales to manage thousands of VMs and hundreds of hosts.
- Offers linked mode for centralized control of distributed environments.

vCenter's comprehensive feature set makes it an essential platform for managing modern virtualized data centers, ensuring performance, reliability, and scalability.

1.5 vSphere Network

vSphere networking is a core component enabling communication between VMs, ESXi hosts, and external networks, providing flexibility, security, and performance optimization.

1.5.1 How Does a Standard vSwitch Work in VMware ESXi?

- **Standard vSwitch (vSS) in VMware ESXi** is a fundamental, host-level networking component enabling communication for VMs and host services both internally and externally. While straightforward and configured independently on each ESXi host, its lack of centralized management makes it less ideal for large-scale environments.
- **Key aspects of vSS include:** its ease of configuration via ESXi/vSphere Client, making it suitable for small to medium setups; support for VLAN segmentation, which enhances security and efficiency by isolating traffic; and basic load balancing with NIC teaming, improving performance and providing redundancy.
- **The architecture comprises:** Port Groups that define network policies for VMs; VMkernel Adapters handling specialized traffic like vMotion, storage communication, and host management; and Uplink Adapters (pNICs) connecting the virtual switch to the physical network.

- **Limitations of vSS are primarily:** the need for per-host configuration due to the absence of centralized management, which can be inefficient in larger deployments; and the lack of advanced networking features such as NetFlow, Port Mirroring, and traffic filtering, crucial for detailed network monitoring and security.
- **In conclusion,** vSS is a reliable and simple networking solution for virtual environments with basic requirements. However, for enterprises demanding centralized control, automation, and advanced networking capabilities, the vSphere Distributed Switch (vDS) is a more appropriate choice.

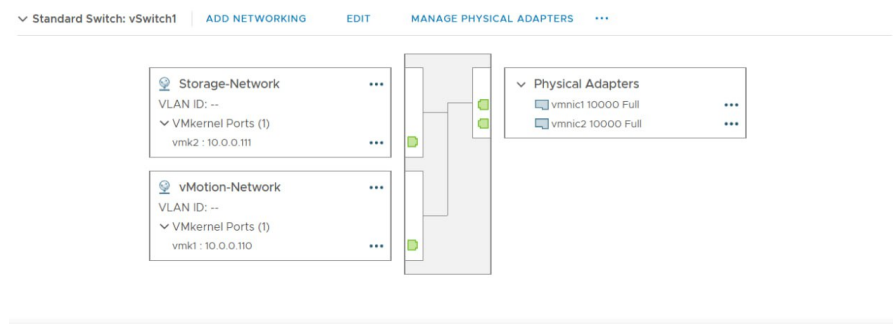


Figure 1.2: Your caption here

1.5.2 How Does a Distributed Switch Work in VMware ESXi?

- **A vSphere Distributed Switch (vDS)** is a networking solution designed for centralized and scalable management of network configurations across multiple ESXi hosts, contrasting with the per-host configuration of a vSS. Managed centrally through vCenter Server, vDS enables consistent network policies, improved resource utilization, and simplified administration, particularly in larger environments.
- **Key advantages of vDS include:** its centralized management, where network settings and policies are defined once in vCenter and applied uniformly, ensuring consistency and reducing management complexity; and the provision of advanced networking features like traffic filtering, NetFlow, Port Mirroring, and Network I/O Control (NIOC), which enhance performance, security, and monitoring. Furthermore, Private VLANs (PVLANS) can isolate VM traffic while sharing physical uplinks, improving security and network segmentation.

1.5.3 What Are Port Groups in vSphere?

Port Groups are logical groupings of virtual network ports on a vSwitch that serve as a bridge between VMs and the physical or virtual network infrastructure. They play a crucial role in defining how VMs interact with the network by establishing rules and policies that control network traffic flow.

One of the primary functions of Port Groups is to enable VMs to connect to the network efficiently. They define important network policies, including VLAN tagging, security settings, and NIC teaming, ensuring that network traffic is properly segmented and optimized. These groups can be found on both vSS and vDS, allowing flexibility in different network configurations..

1.5.4 What Are VMkernel Adapters?

- A **VMkernel Adapter (vmk)** is a specialized network interface within an ESXi host, dedicated to facilitating host-level networking services. Unlike virtual machine network adapters used by VMs, VMkernel Adapters are exclusively utilized by the ESXi host for its internal communication and various management tasks, playing a crucial role in ensuring seamless interaction between the host and network-based services.
- The **primary function of a VMkernel Adapter** is to enable network connectivity for essential host services, including management (connecting to vCenter, SSH, etc.), vMotion (live VM migration), storage (iSCSI/NFS, vSAN), Fault Tolerance (FT logging), and provisioning (cloning, snapshots). Each VMkernel Adapter is assigned an IP address, allowing the ESXi host to communicate over the network, and these adapters can be configured on both Standard vSwitches (vSS) and Distributed Switches (vDS), depending on the virtual infrastructure's design and requirements.
- **Different types of VMkernel services** can be configured based on their specific purpose. Management Traffic allows administrators to connect to ESXi hosts for control. vMotion Traffic enables the live migration of VMs without downtime. Provisioning Traffic is used for VM cloning and snapshot operations. vSAN Traffic facilitates communication with VMware's software-defined storage. FT Logging ensures real-time replication for continuous availability. Finally, iSCSI/NFS Storage Traffic enables communication with external storage solutions.

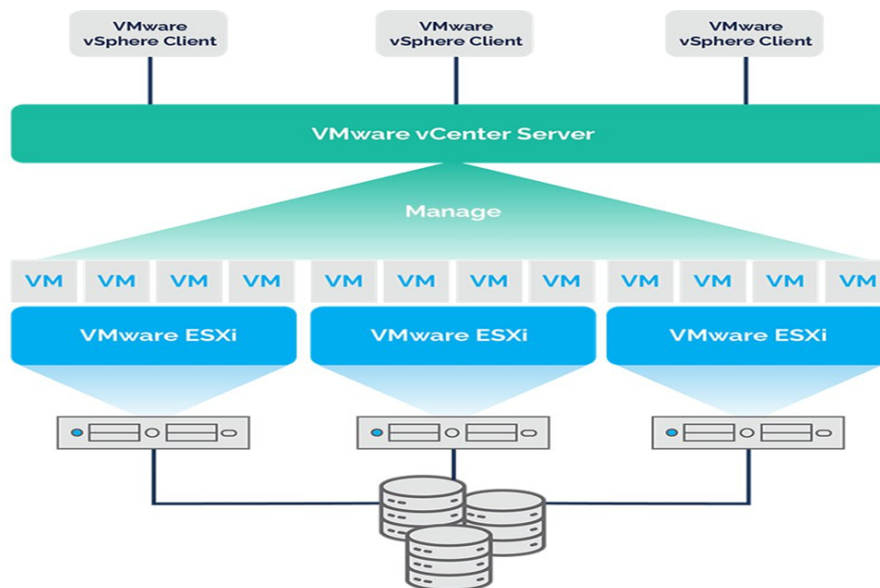


Figure 1.3: Vsphere Network

Virtual Storage

Virtual storage is a core concept in VMware vSphere, designed to abstract physical storage devices into logical units that can be easily managed and allocated. This abstraction allows multiple virtual machines (VMs) to share the same storage pool efficiently. In critical environments like hospitals, where systems such as ICU dashboards, pharmacy databases, and patient record services demand high-speed, reliable access to data, virtual storage becomes a fundamental building block. By separating the physical characteristics of storage from how it is logically presented, virtual storage improves scalability, flexibility, and performance across a virtualized infrastructure.

In vSphere, virtual storage is implemented through datastores, which are logical containers for VM files such as virtual disk files (.vmdk), configuration files (.vmx), logs, ISO images, and snapshots. These datastores are built on top of different physical storage technologies and protocols, including block-level and file-level storage systems.

Types of Virtual Storage

Direct-Attached Storage (DAS):

DAS consists of local physical disks (HDDs or SSDs) directly attached to the ESXi host through interfaces such as SATA, SAS, or NVMe. DAS is the simplest and most cost-effective storage type. However, its major limitation is that the storage is accessible only by the host it is attached to. This makes DAS unsuitable for clustered environments or advanced features such as vMotion or High Availability (HA).

Use Case: DAS is commonly used for small environments, testing, or non-critical workloads that do not require shared access.

Network-Attached Storage (NAS):

NAS provides file-level access to shared storage over a standard TCP/IP network using the Network File System (NFS) protocol. It is easy to configure, does not require specialized storage hardware, and is typically used for non-performance-intensive tasks like hosting ISO images, VM templates, and backups. Since it offers shared access, NAS enables multiple ESXi hosts to read and write to the same datastore, although at a lower performance level compared to block-based storage.

Use Case: NAS is ideal for environments that prioritize ease of deployment and shared access without requiring the high performance of SAN.

Storage Area Network (SAN):

SAN is a high-performance, block-level storage system that connects to ESXi hosts via specialized protocols. It delivers fast, low-latency storage and supports enterprise-grade workloads. SAN uses Logical Unit Numbers (LUNs) to allocate storage volumes to hosts. These LUNs are formatted using VMFS (VMware File System) and presented to multiple hosts, enabling concurrent access to VMs. SAN supports features like vMotion, clustering, and Storage DRS.

Use Case: SAN is commonly deployed in enterprise hospital systems for critical services like medical databases and patient monitoring platforms.

Datastore Types

VMFS (VMware File System):

VMFS is a clustered file system used to format block-based storage such as SAN or DAS. It enables multiple ESXi hosts to access the same datastore concurrently and supports features such as vMotion, snapshots, and thin provisioning. VMFS is optimized for storing large numbers of virtual machines and allows secure file locking to prevent corruption.

NFS (Network File System):

NFS datastores are mounted directly from NAS servers over TCP/IP. They provide file-level access and are managed by the storage device itself. While not as fast as VMFS, NFS supports snapshots and space-saving technologies like deduplication, making it useful for secondary storage or less demanding workloads.

In conclusion, virtual storage is a foundational part of vSphere that allows for dynamic, centralized storage management across a virtualized environment. By selecting the right combination of storage type, transport protocol, and datastore format, administrators can design systems that meet performance, availability, and scalability requirements. In healthcare environments like hospitals, where reliability and speed are essential, SAN storage with VMFS formatting is often the best solution for critical workloads, while NAS can serve effectively for backups and templates.

1.6 How Are VM Templates Created in ESXi 7 and vCenter?

VM templates play a crucial role in efficient virtual machine deployment within ESXi 7 and vCenter. They allow administrators to create standardized, pre-configured virtual machines, reducing manual setup time and ensuring consistency across deployments. Understanding VM templates and how they differ from VM clones is essential. While a VM clone is an exact copy of a virtual machine at a specific moment, a VM template serves as a master image that can be reused to create multiple VMs without modification.

1.6.1 Overview

- To create a VM template, an administrator must first configure a source VM with essential updates, drivers, and security settings. The VM is then converted into a template within vCenter, ensuring it is locked from accidental changes. Best practices include applying OS patches, installing necessary applications, and optimizing system settings before converting the VM to a template.
- Once a template is created, multiple virtual machines can be deployed from it using vCenter. During deployment, administrators can customize VM settings, such as CPU, memory, and network configurations, allowing flexibility while maintaining standardization. This streamlined process significantly accelerates the provisioning of VMs in enterprise environments.
- Since software updates and security patches are constantly released, VM templates need regular updates to remain useful. vCenter allows administrators to convert a template back into a virtual machine, apply necessary changes, and then reconvert it into an updated template. This ensures that newly deployed VMs remain secure and up to date.

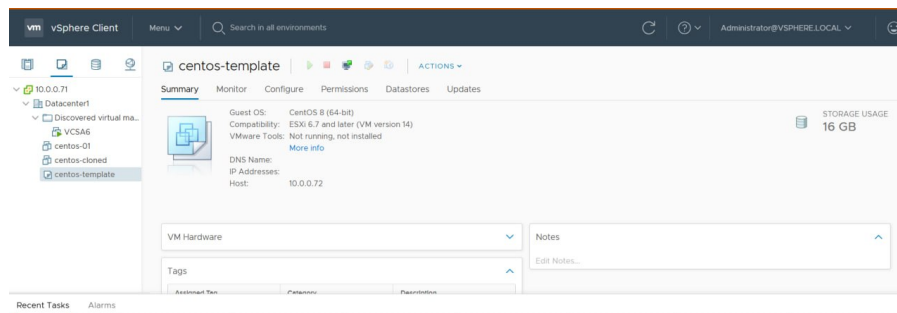


Figure 1.4: Template

1.7 VM Migration in vSphere

VM migration in vSphere is a fundamental capability that allows VMs to be transferred across hosts, clusters, and storage environments with minimal or no downtime. This feature is essential for optimizing resource utilization, maintaining HA, and facilitating hardware maintenance without disrupting services. VMware provides various migration methods tailored to different scenarios, including Cold Migration, vMotion, Storage vMotion (sVmotion), and Shared Nothing vMotion.

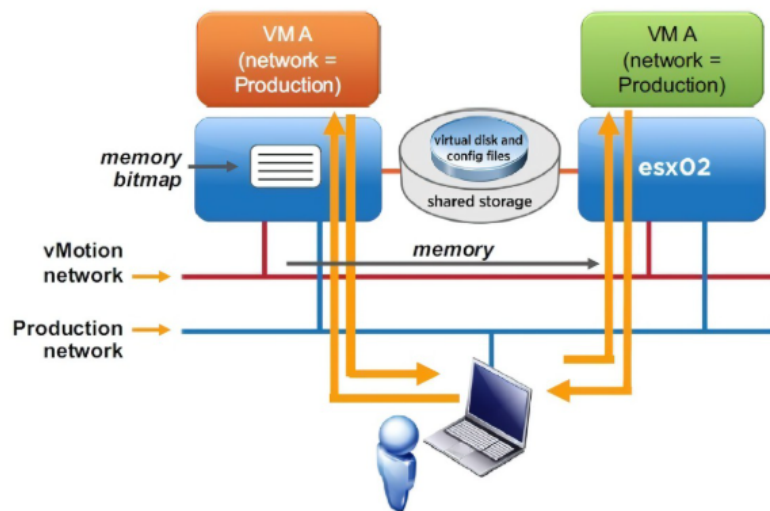


Figure 1.5: Vm Migration

1.7.1 Cold Migration

Cold Migration involves moving a virtual machine while it is powered off. This method is commonly used when planned downtime is acceptable, such as during scheduled maintenance, infrastructure upgrades, or migrations between vCenters. Cold Migration allows VMs to be transferred between different hosts, clusters, or storage environments, including from local disks to shared storage.

Since the VM is not running, there is no impact on performance, but the migration process is slower compared to live migration methods. It is particularly useful for minimizing risk when relocating workloads between datacenters or consolidating resources.

1.7.2 vMotion

vMotion is VMware's live migration technology that enables the seamless transfer of a running VM from one ESXi host to another without downtime. This is achieved by copying the VM's active memory state, CPU execution context, and network connections while keeping the virtual disk files on shared storage. The process happens in real-time, ensuring minimal disruption to applications and users.

1.7.3 Storage vMotion

Storage vMotion allows the migration of a VM's disk files between different datastores while the VM remains powered on. This feature is useful for optimizing storage resources, upgrading storage infrastructure, or reallocating VMs to more cost-effective storage solutions. Storage vMotion leverages Changed Block Tracking (CBT) to copy disk data incrementally, ensuring minimal impact on VM performance.

This method supports a wide range of storage types, including local storage, NFS, iSCSI, Fiber Channel, and VMware vSAN. To perform a successful Storage vMotion, the ESXi host must have access to both the source and destination datastores, and the environment must be licensed with vSphere Enterprise or higher. Common use cases include migrating VMs to faster storage, consolidating workloads onto a new storage array, and reducing costs by moving less critical VMs to lower-tier storage.

1.7.4 Shared Nothing vMotion

Shared Nothing vMotion expands traditional vMotion capabilities by allowing VMs to migrate between hosts with different storage. Unlike standard vMotion, which requires shared storage, this method combines vMotion for memory and CPU state migration with Storage vMotion for disk migration. The process starts by copying the VM's disk files from the source to the destination host's storage, followed by the transfer of memory and network connections.

This method is particularly beneficial for organizations transitioning from local storage to a shared storage infrastructure or for migrating workloads between geographically dispersed datacenters. Shared Nothing vMotion eliminates the need for downtime, making it a valuable tool for workload mobility and infrastructure consolidation.

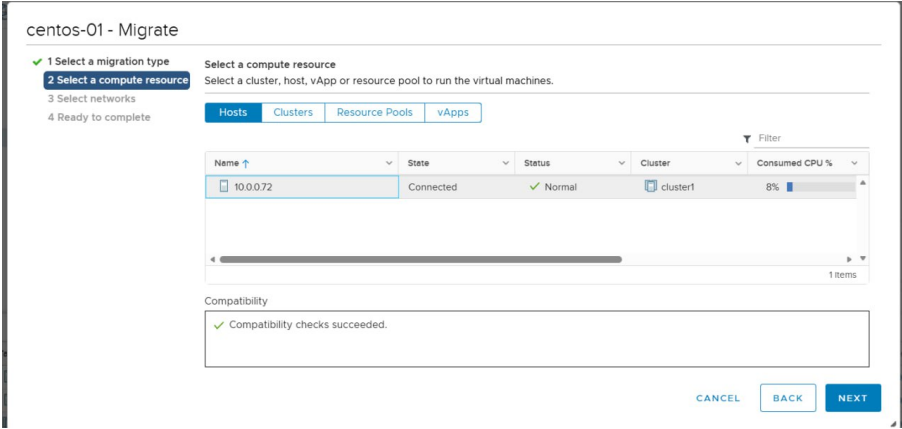


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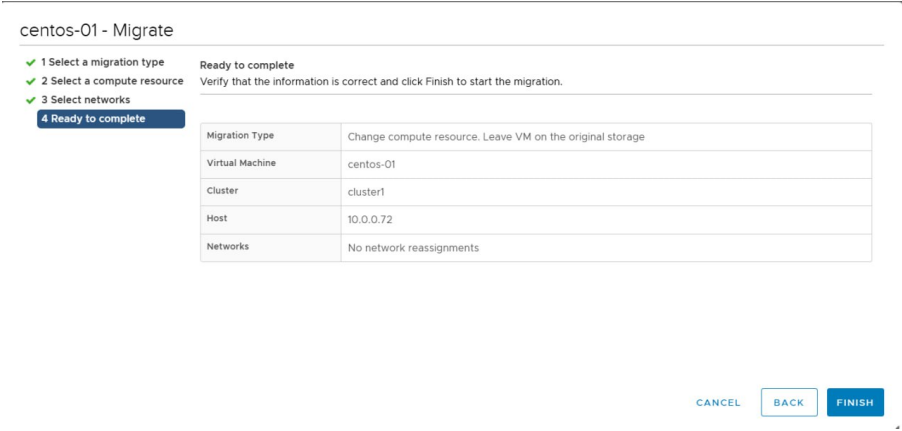


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1.8 vSphere Clusters

A vSphere Cluster is a logical grouping of multiple ESXi hosts that function as a single computing resource to provide HA, load balancing, and scalability for virtual machines. By pooling resources such as CPU, memory, and storage, a vSphere Cluster ensures that workloads are efficiently distributed and can automatically recover from hardware failures

1.8.1 Key Components of a vSphere Cluster

- ESXi hosts form the foundation of a vSphere cluster, providing the necessary compute power and storage for VMs. Each host contributes its resources to the cluster, enabling VMs to operate seamlessly across multiple hosts. By pooling resources together, vSphere clusters enhance performance, scalability, and HA for virtualized workloads.
- A key advantage of a vSphere cluster is resource pooling, which aggregates the CPU and memory resources of all hosts into a unified pool. VMs within the cluster dynamically consume resources based on their workload demands and configured settings. This dynamic allocation ensures optimal performance, prevents resource wastage, and enables administrators to efficiently manage workloads across the cluster.
- To enable advanced vSphere features like vMotion and HA, shared storage is required. All ESXi hosts in the cluster typically connect to a shared storage solution such as a SAN, VMware vSAN, or Network Attached Storage (NAS). Shared storage ensures that VMs can be migrated between hosts without downtime and that critical workloads remain available in the event of a host failure.
- Networking in a vSphere cluster must be configured consistently across all hosts to ensure seamless communication and VM mobility. A well structured virtual network allows VMs to interact efficiently while maintaining security and performance. The use of vDS simplifies network management by providing a centralized way to configure and control network settings across multiple ESXi hosts. Distributed switches also enable features like traffic shaping, network isolation, and improved monitoring, further enhancing the cluster's networking capabilities.

1.8.2 vSphere Cluster Architecture

- A vSphere Cluster operates as a unified entity where multiple ESXi hosts collaborate to provide a highly available and scalable environment. This architecture enables resource sharing, workload balancing, and advanced virtualization features such as vMotion and HA. The design of a vSphere cluster ensures efficient management and optimal utilization of compute and storage resources.
- A vSphere Cluster consists of two or more ESXi hosts, which work together to distribute workloads and enhance system resilience. To ensure smooth operations, hosts within a cluster should have similar CPU and memory resources to maintain consistent performance across virtual machines. The more uniform the hardware configurations, the better the cluster can function without compatibility issues.

- vSphere clusters are designed for scalability, allowing organizations to expand their infrastructure as needed. The cluster can scale horizontally by adding more ESXi hosts, increasing the available CPU, memory, and storage resources. The total number of virtual machines that can run in a cluster depends on the aggregate resources of the hosts. VMware supports large-scale clusters, with the maximum number of hosts per cluster reaching up to 96, depending on the vSphere version.

1.9 VMware vSphere High Availability

VMware vSphere High Availability (HA) is a feature that ensures business continuity by automatically restarting VMs on available hosts when a failure occurs. This minimizes downtime and reduces manual intervention in the event of hardware or host failures. vSphere HA is particularly useful in environments that require high uptime and minimal service disruption.

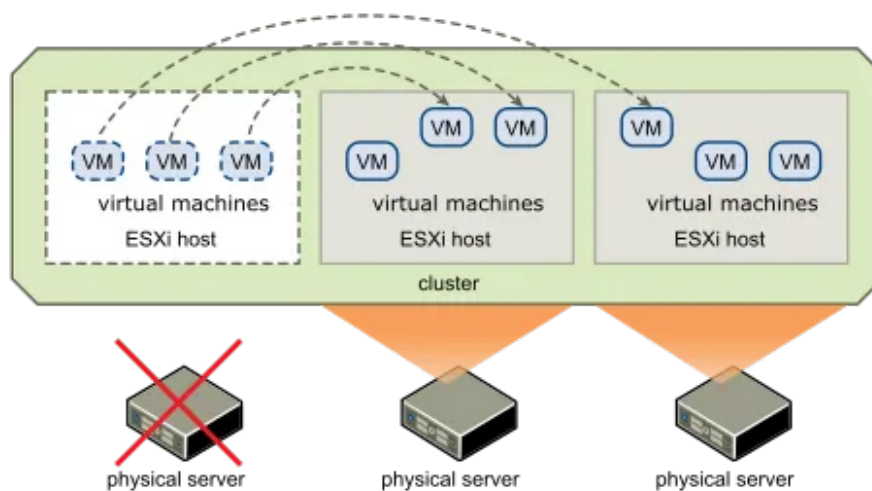


Figure 1.8: High Availability

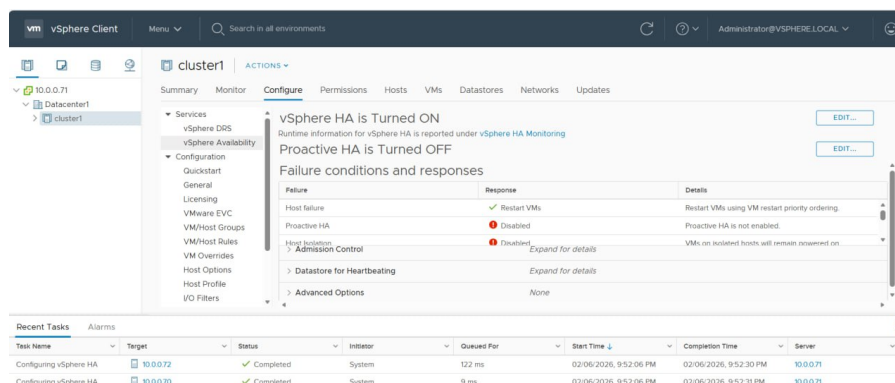


Figure 1.9:

1.9.1 Key Components of vSphere HA

- **HA Agent (Fault Domain Manager - FDM):** each ESXi host in an HA cluster runs the Fault Domain Manager (FDM) agent, which is responsible for monitoring the health of hosts and detecting failures. The FDM agent communicates with vCenter Server, reporting any detected issues and initiating automated failover processes when necessary.

If an ESXi host fails, the FDM master host takes control, ensuring that affected virtual machines are restarted on available hosts.

Communication between FDM agents happens through the Management VMkernel Service, which keeps all hosts in sync and ensures that configuration updates are applied cluster-wide. When changes occur within the cluster, vCenter pushes updated configurations to all hosts, storing them in the local datastore of each host. This approach enhances reliability and ensures that vSphere HA can quickly react to unexpected failures.

- **Master and Slave Hosts:** In a vSphere HA cluster, ESXi hosts operate in either a master or slave role. The Master Host is responsible for monitoring the status of all other hosts and managing failover operations when failures occur. If an ESXi host in the cluster becomes unavailable, the master host ensures that its virtual machines are restarted on other available hosts. Although vCenter Server serves as a management interface, it does not execute HA operations directly—the master host handles these tasks independently.

Slave hosts, on the other hand, report their status to the master host. If a host failure is detected, the master coordinates VM restarts across the cluster. In the event that the master host itself fails, an automated election process is triggered to select a new master, ensuring uninterrupted HA functionality without requiring manual intervention.

- **Master Host Election:** When vSphere HA is enabled, an election process takes place to determine which ESXi host will assume the role of Master Host. This ensures that the cluster remains operational even in the event of failures. The election process is initiated when vSphere HA is first enabled or when the current master host becomes unavailable.
- **What Happens When the Master Host Fails?** If the master host becomes unreachable, the cluster must recover quickly to maintain high availability. The first step in recovery is failure detection. Slave hosts attempt to communicate with the master using network heartbeats. If no response is received within 15 seconds, the slaves assume that the master has failed.

To distinguish between a total host failure and a network partition, the slave hosts also check for datastore heartbeats to determine whether the failure is due to connectivity issues or an actual host outage.

Once a failure is confirmed, the new master election process begins. The cluster follows the same selection criteria used in the initial election, ensuring that a new master is chosen within 30 seconds to 1 minute. After the election, the new master takes over cluster control, monitors all slave hosts, and ensures that virtual machines are properly restarted if necessary. HA reconfigures itself to restore full functionality, preventing disruptions to running workloads.

1.9.2 Host Failure Types

When a host is disconnected from a vSphere HA cluster, the virtual machines registered to that host become unprotected. This means that if the host experiences a failure, vSphere HA cannot automatically restart the affected virtual machines on another host. Proper configuration and monitoring of HA are essential to ensure continuous protection for workloads in a cluster environment.

Host Failure Types in vSphere HA

In a vSphere HA cluster, the primary host is responsible for detecting failures of secondary hosts and taking the necessary recovery actions.

Host failures can be categorized into three main types:

- **Host Failure:** This occurs when a host stops functioning completely. The affected virtual machines must be restarted on another available host.
- **Host Isolation:** A host is still running but becomes disconnected from the management network. In this case, the host cannot communicate with other HA agents.
- **Host Partition:** A host loses connectivity with the primary host, creating a separate partition. Although the host is still operational, it cannot communicate with the rest of the cluster.

Host Network Isolation in vSphere HA

- A host is considered isolated when it is still running but no longer detects any vSphere HA agent traffic on the management network.
- To confirm isolation, the host attempts to ping the cluster isolation addresses configured for the HA cluster. If these pings also fail, the host declares itself as isolated.
- When a host becomes isolated, the HA cluster takes appropriate actions based on the configured host isolation response policy. These responses may include:
 - Powering off or restarting affected virtual machines on other available hosts.
 - Leaving virtual machines running on the isolated host (if configured to do so).
- Proper network redundancy and isolation response settings are crucial to mitigating the risks associated with host isolation.

Datastore Heart beating in vSphere HA

- To improve failure detection in cases where network communication is disrupted, vSphere HA uses datastore heart beating as a secondary mechanism. This feature helps distinguish between a true host failure and a network partition or isolation event.
- How Datastore Heart beating Works:
 - **Communication Failure Detection:** If the primary host stops receiving network heartbeats from a secondary host, it assumes a potential failure and begins additional checks. This can occur due to network partitioning, isolation, or an actual host failure.
 - **Datastore Heartbeat Check:** The primary host checks whether the secondary host is still writing heartbeat signals to a shared datastore. These signals indicate that the secondary host is operational despite network issues.
 - **Failure Determination:** If the secondary host is no longer writing heartbeats to the datastore, the primary host considers it to have failed.

- **VM Restart:** Once a failure is confirmed, vSphere HA automatically restarts the virtual machines that were running on the failed host, ensuring continued service availability.
- By leveraging datastore heart beating, vSphere HA enhances its ability to differentiate between network failures and actual host failures, reducing unnecessary VM restarts and improving cluster stability.
- This mechanism plays a vital role in maintaining high availability of virtualized workloads, even in the face of network disruptions.

1.9.3 Failure Scenarios

This subsection explains the behavior of the system **before and after a failure** to illustrate how High Availability works.

Before Failure: Before a failure occurs, HA systems are configured to monitor resources and maintain redundancy. Key points include:

- Multiple nodes or servers are available to handle workloads.
- Regular health checks are performed to detect potential issues.
- Data replication ensures consistency across nodes.
- Failover mechanisms are tested and ready to activate if needed.

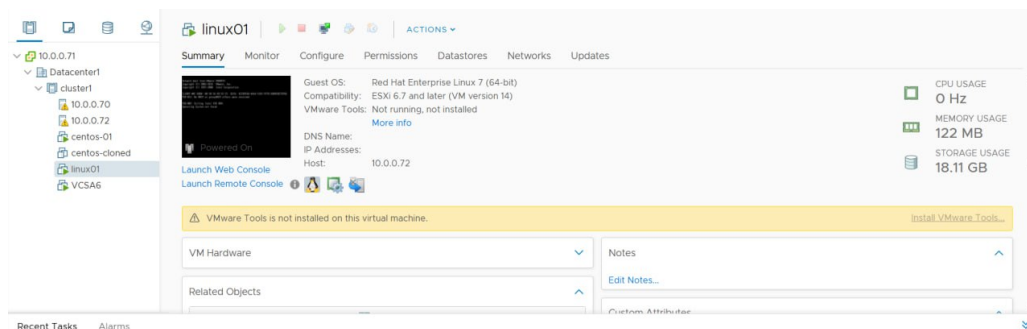


Figure 1.10: High Availability: Before Failure

After Failure: After a failure occurs, HA systems respond automatically to minimize service disruption. Key points include:

- Traffic is redirected to available nodes or servers.
- Failed components are isolated to prevent impact on other services.
- Recovery processes restore functionality and data integrity.
- Alerts are sent to administrators for further action or maintenance.

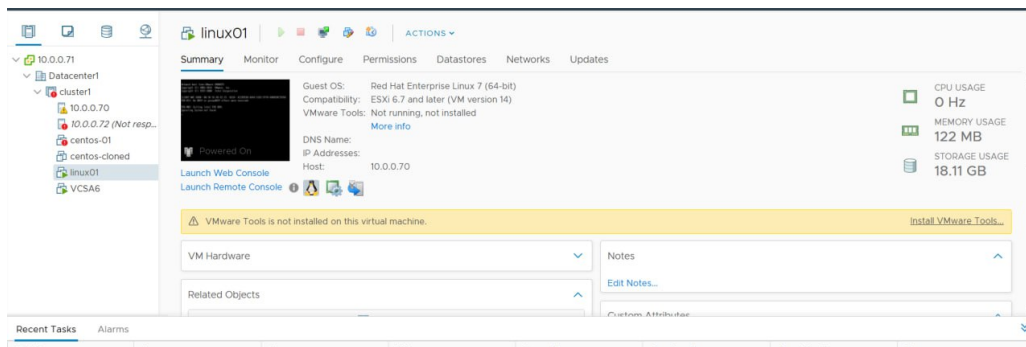


Figure 1.11: High Availability: after Failure

1.9.4 vSphere HA Admission Control

vSphere HA uses admission control to ensure that sufficient resources are reserved for virtual machine recovery in case of a host failure. This mechanism enforces resource constraints, preventing actions that might violate failover guarantees, such as powering on a virtual machine, migrating workloads, or increasing CPU and memory reservations beyond the available failover capacity.

The primary goal of admission control is to ensure that the cluster can tolerate a specified number of host failures while maintaining virtual machine availability. Without this safeguard, a host failure might cause critical virtual machines to be unavailable, disrupting operations.

The host failover capacity in vSphere HA can be configured in three ways: Cluster Resource Percentage, Slot Policy, and Dedicated Failover Hosts. Each method determines how resources are allocated to guarantee failover capacity.

The host failover capacity in vSphere HA can be configured in three ways:

- **Cluster Resource Percentage:** This method reserves a specific percentage of the cluster's total CPU and memory resources for failover. When admission control is enforced, vSphere HA calculates the total resource requirements of all powered-on virtual machines and compares them against the total host resources available. The system then determines the Current CPU and Memory Failover Capacity, ensuring that it does not fall below the Configured Failover Capacity set by the user. If the available capacity is insufficient, admission control prevents further operations that could impact failover readiness.

vSphere HA uses actual resource reservations to compute failover capacity. If a virtual machine does not have an explicit reservation, a default value of 0MB memory and 32MHz CPU is applied.

Additionally, the cluster must have at least two HA-enabled hosts to function correctly. Even if resource percentages indicate sufficient capacity, failover cannot occur if only one host remains.

- **Slot Policy:** The Slot Policy method divides the cluster's resources into logical units called slots, which represent a standardized measure of CPU and memory allocation for virtual machines. vSphere HA calculates the slot size based on the largest CPU and memory reservation of any powered-on virtual machine in the cluster. Each ESXi host can accommodate a certain number of slots, and the total number of available slots determines the cluster's failover capacity.

When a host failure occurs, vSphere HA ensures that enough slots remain to restart all virtual machines from the failed host. If the number of remaining slots falls below the Configured Failover Capacity, admission control prevents additional workloads from being powered on.

However, the Slot Policy approach has limitations, as it may overestimate or underestimate failover capacity, leading to inefficient resource utilization.

- **Dedicated Failover Hosts:** With the Dedicated Failover Hosts method, specific ESXi hosts are reserved exclusively for failover. When a host fails, vSphere HA prioritizes restarting virtual machines on these designated failover hosts. If these hosts are unavailable or lack sufficient resources, vSphere HA attempts to restart the workloads on other available hosts in the cluster.

To ensure that reserved capacity remains available, failover hosts cannot be used for regular workloads. Virtual machines cannot be manually powered on or migrated to these hosts using vMotion, and DRS does not use failover hosts for load balancing.

Additionally, if multiple failover hosts are designated, DRS does not enforce VM-VM affinity rules for virtual machines running on them.

Edit Cluster Settings
cluster1

vSphere HA

Failures and responses
Admission Control
Heartbeat Datastores
Advanced Options

Admission control is a policy used by vSphere HA to ensure failover capacity within a cluster. Raising the number of potential host failures will increase the availability constraints and capacity reserved.

Host failures cluster tolerates
1
Maximum is one less than number of hosts in cluster.

Define host failover capacity by
Cluster resource Percentage

☐ Override calculated failover capacity.

Reserved failover CPU capacity:
0
% CPU

Reserved failover Memory capacity:
0
% Memory

Performance degradation VMs tolerate
100
%
Percentage of performance degradation the VMs in the cluster are allowed to tolerate during a failure. 0% - Raises a warning if there is insufficient failover capacity to guarantee the same performance after VMs restart. 100% - Warning is disabled.

CANCEL
OK

Figure 1.12: admission control

1.10 vSphere Distributed Resource Scheduler

1.10.1 Overview

- **vSphere Distributed Resource Scheduler (vDRS)** is a powerful vCenter feature that ensures efficient workload distribution across ESXi hosts in a cluster by dynamically balancing VMs based on CPU and memory availability.
- **Leveraging vMotion**, vDRS optimizes performance by preventing resource contention and enhancing infrastructure efficiency through continuous, real-time monitoring and adjustments of workload placement.
- **A core function of vDRS is load balancing**, where it continuously tracks CPU and memory usage across hosts and automatically initiates vMotion to migrate VMs from overloaded to underutilized hosts.
- **This automated process ensures even workload distribution**, preventing performance bottlenecks and maintaining smooth operations without administrator intervention.

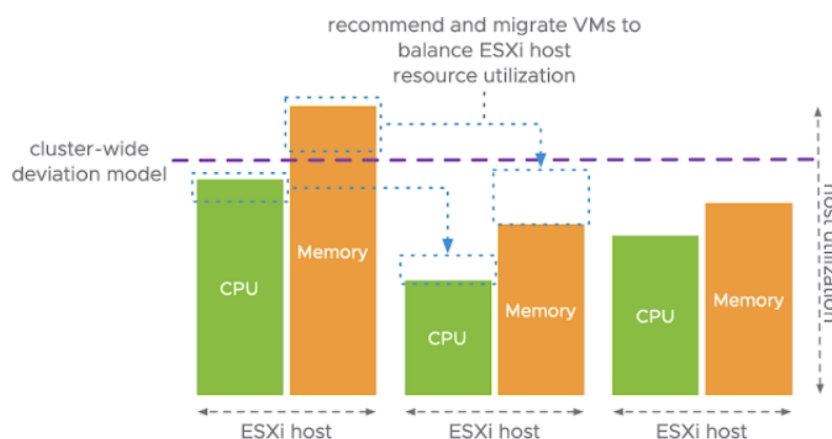


Figure 1.13: Vdrs Overview

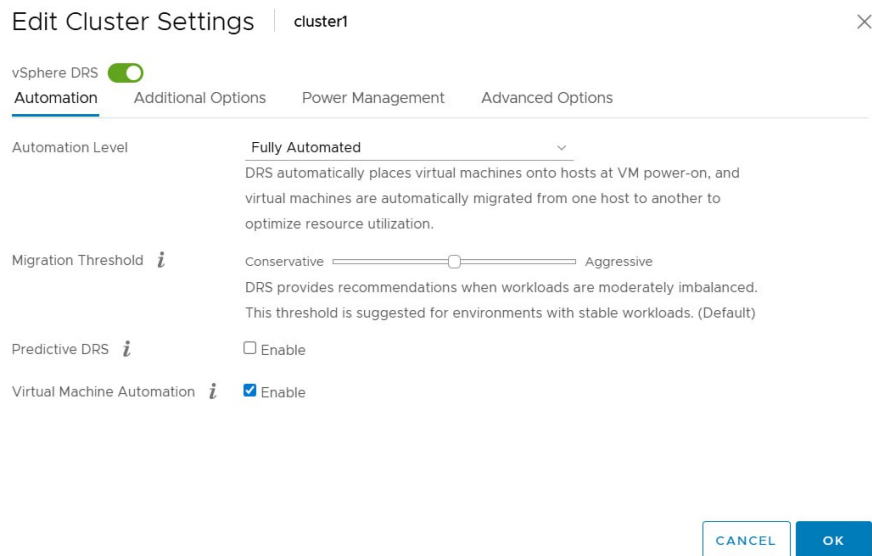


Figure 1.14:

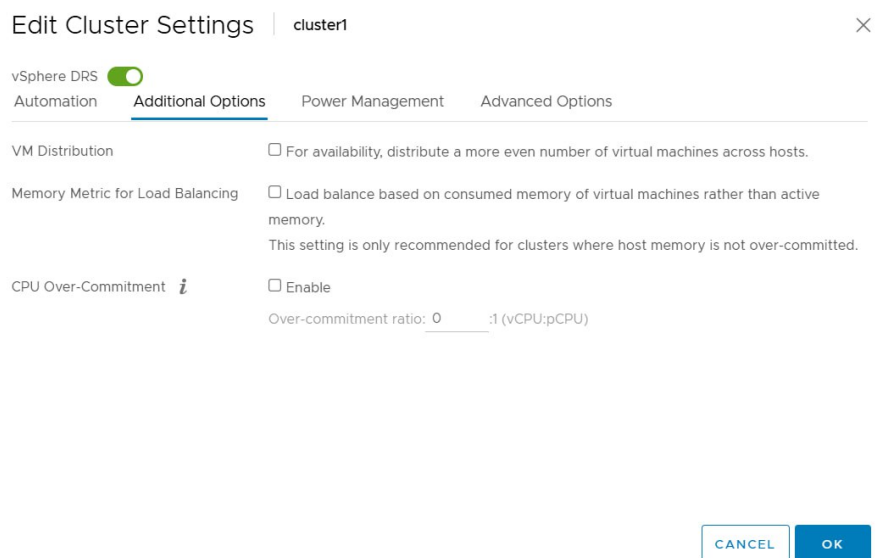


Figure 1.15:

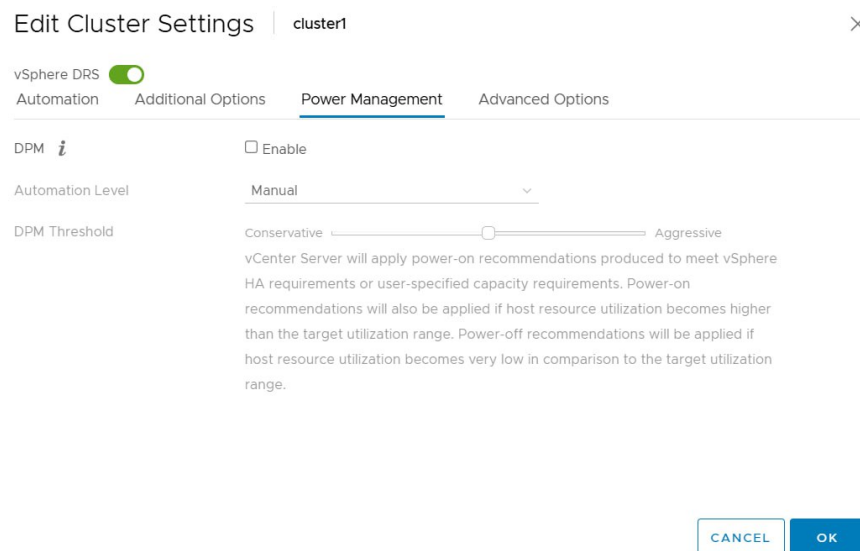


Figure 1.16:

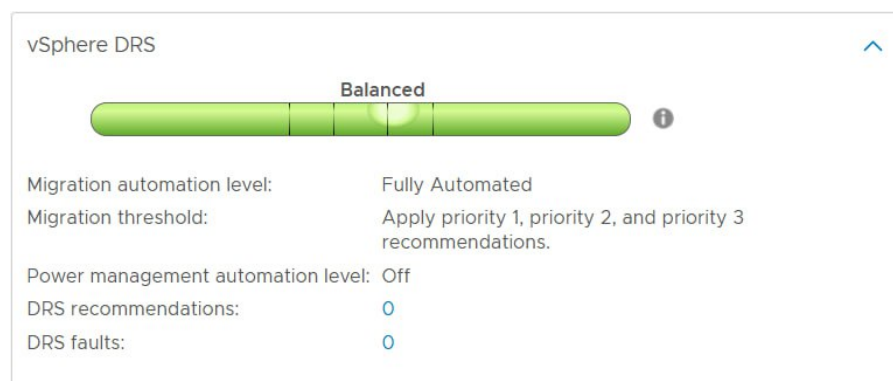


Figure 1.17:

1.10.2 VM Placement and Resource Allocation

- **Intelligent Workload Distribution:** vDRS ensures efficient workload distribution by intelligently placing new virtual machines and managing resource allocation within a vSphere cluster.
- **Automated Host Selection:** Upon VM power-on, vDRS analyzes real-time CPU and memory availability to determine the most suitable ESXi host. This automated selection prevents initial resource imbalances, ensuring optimal performance and stability within the cluster.
- **Dynamic VM Placement:** By dynamically placing virtual machines based on available resources, vDRS enhances overall efficiency and minimizes the occurrence of performance bottlenecks.
- **Resource Pools for Prioritization:** DRS allows administrators to create resource pools, defining CPU and memory allocation based on organizational priorities. These pools enable the assignment of specific shares, limits, and reservations to workloads, guaranteeing that critical applications receive the necessary resources.

1.10.3 Distributed Power Management

vDRS includes DPM an optional feature designed to optimize energy consumption. During periods of low resource demand, DPM consolidates workloads and powers down idle hosts to reduce energy costs. When resource demand increases, the powered-down hosts are automatically reactivated to accommodate the additional load. This capability makes vSphere clusters more energy-efficient, reducing operational costs while maintaining high availability.

1.10.4 Automation Levels

- **Manual Mode:** vDRS provides recommendations for VM migration, requiring manual administrator approval for each migration before execution.
- **Partially Automated Mode:** vDRS automatically selects the optimal ESXi host for new VM placements, but administrator approval is necessary for VM migrations triggered by load balancing needs.
- **Fully Automated Mode:** vDRS autonomously manages both the initial placement of new VMs and subsequent live migrations without any administrator intervention, ensuring seamless optimization and overall efficiency.

Migration Sensitivity:

- **Lower Setting (1):** Migrations occur only when significant resource imbalances are detected, minimizing the frequency of VM movements.
- **Higher Setting (5):** Increases the frequency of migrations, prompting responses to even minor variations in workload distribution, which is particularly advantageous for high-performance environments demanding consistent optimization.

1.10.5 Advanced Capabilities and Benefits of vDRS

One of its key features is vDRS with Storage DRS, which optimizes both compute and storage resources by ensuring that VMs are placed on the best combination of host and datastore. This integration improves performance and prevents storage bottlenecks. Additionally, vDRS Rules allow administrators to define affinity and anti-affinity rules, ensuring that critical workloads either stay on designated hosts or are distributed across different hosts to enhance fault tolerance. Another powerful feature is Predictive DRS, which integrates with vRealize Operations to analyze historical usage data and anticipate future resource demands. This proactive approach enables vDRS to migrate VMs before performance issues arise, ensuring better cluster stability and responsiveness.

1.11 Shared Storage and NFS

Shared storage allows multiple systems or servers to access the same storage resources. This is essential in environments like virtualization, clustering, and large-scale data centers, where data consistency and accessibility are critical. Shared storage solutions can include SAN (Storage Area Network), NAS (Network Attached Storage), or cloud-based storage.

1.11.1 Overview of Shared Storage

Shared storage provides a centralized location for storing data that can be accessed by multiple clients. Key advantages include:

- Efficient resource utilization and reduced duplication of data.
- Simplified backup and disaster recovery.
- Improved collaboration among multiple servers or users.

1.11.2 Network File System (NFS)

NFS is a protocol that allows systems to share directories and files over a network. Using NFS, a server can export one or more directories, and clients can mount them as if they were local directories. NFS is widely used in Linux and Unix environments for shared storage because it is simple, reliable, and allows centralized file management.

Key Features of NFS:

- Allows multiple clients to access the same files concurrently.
- Supports different access permissions for users and groups.
- Network transparent: clients interact with remote files as local files.
- Can be integrated with backup, replication, and virtualization solutions.

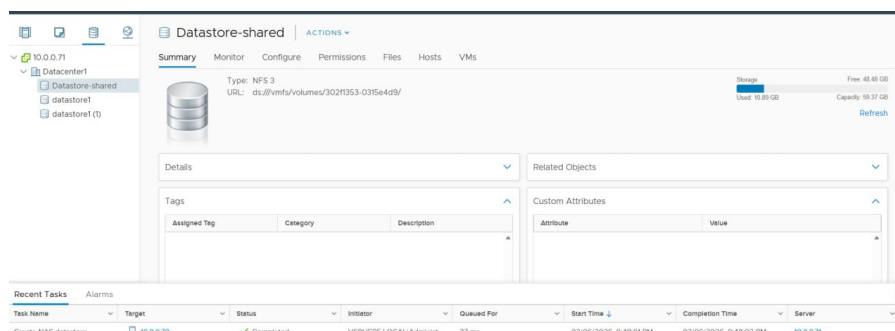


Figure 1.18: NFS Shared Storage Configuration

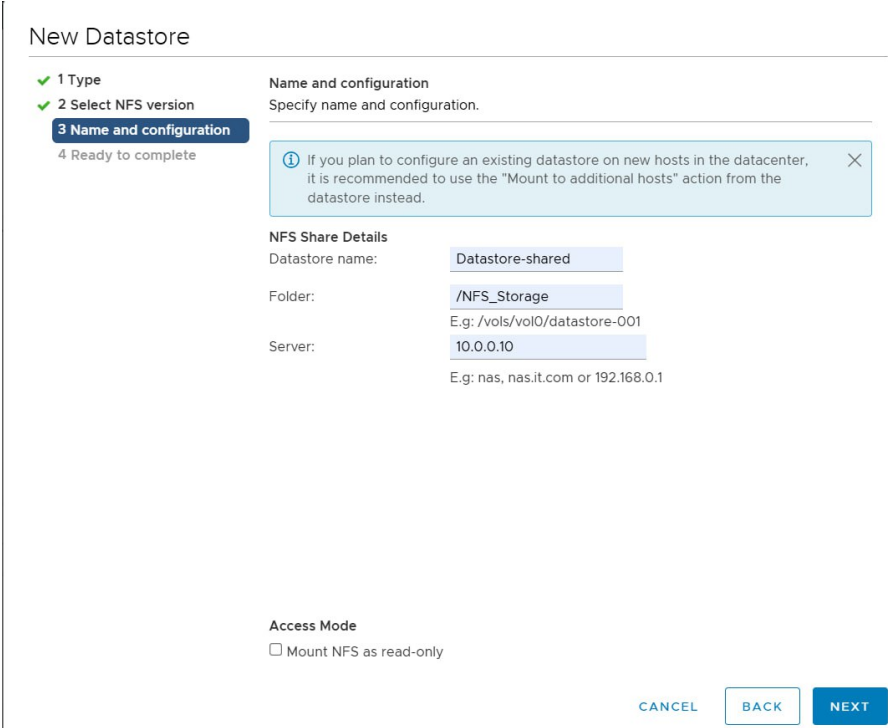


Figure 1.19: NFS Mounting Process

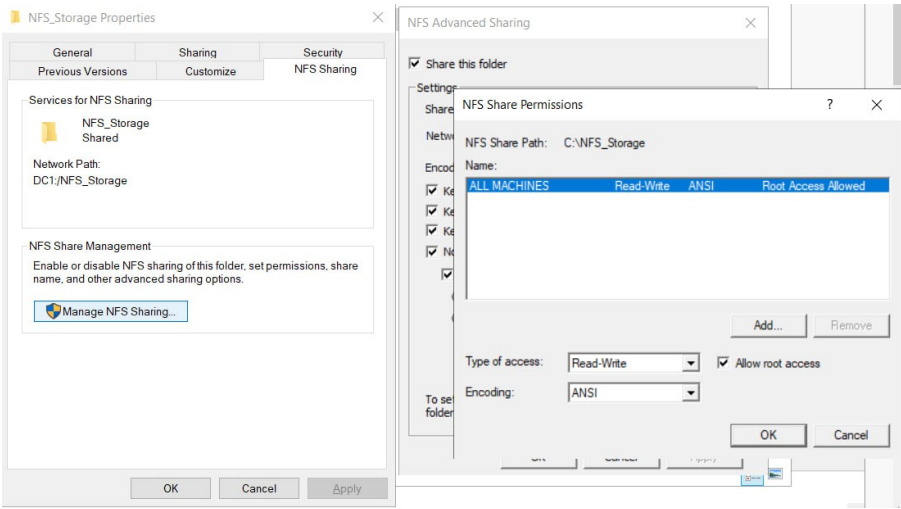


Figure 1.20: NFS Storage Usage in Virtualization

1.12 Content Library

This section contains all the reusable content, assets, and resources used in the project. It may include images, templates, code snippets, or any files needed to replicate the project.

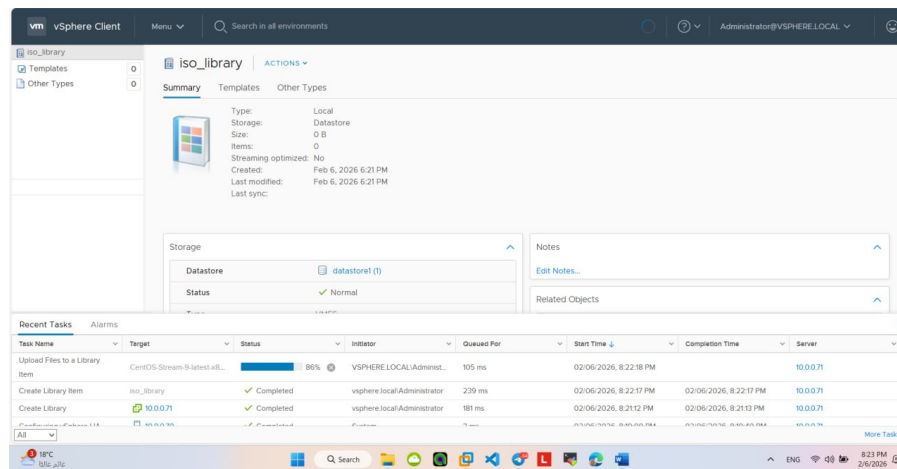


Figure 1.21: Content Library Item 1

Upload Files to a Library	CentOS-Stream-9-latest.x86_64	Completed	VSPHERE.LOCAL\Administ...	105 ms	02/06/2026, 8:22:18 PM	02/06/2026, 8:22:18 PM	10.0.0.71
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Figure 1.22: Content Library Item 2

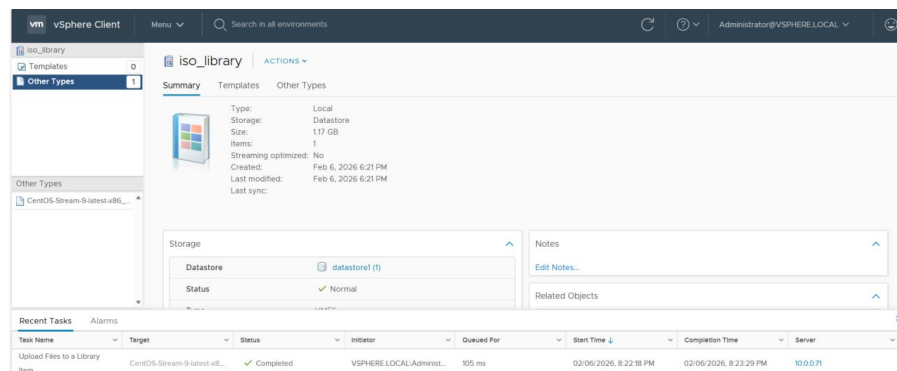


Figure 1.23: Content Library Item 3

1.13 Fault Tolerance

Fault Tolerance (FT) is a mechanism that ensures continuous system operation without interruption, even when a hardware failure occurs. Unlike High Availability, which restores services after a failure, Fault Tolerance prevents downtime by maintaining a continuously running duplicate of a virtual machine.

Fault Tolerance works by creating a primary virtual machine and a secondary virtual machine that runs in parallel on another host. Both machines execute the same operations simultaneously. If the primary host fails, the secondary virtual machine immediately takes over without service interruption or data loss.

1.13.1 Key Characteristics of Fault Tolerance

- Provides zero downtime and zero data loss.
- Uses a secondary VM that mirrors the primary VM in real time.
- Requires shared storage accessible by multiple hosts.
- Automatically switches to the secondary VM if failure occurs.

1.13.2 Fault Tolerance Architecture

The following figure illustrates how Fault Tolerance maintains service continuity using primary and secondary virtual machines.

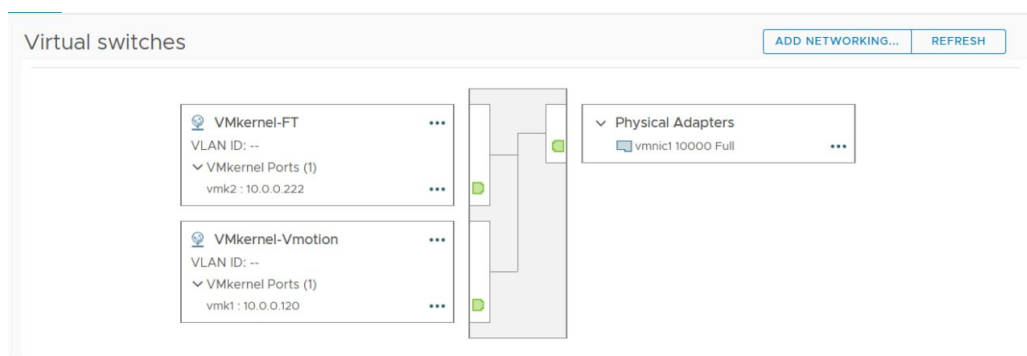


Figure 1.24: Fault Tolerance Architecture

linux01 - Turn On Fault Tolerance

1 Select datastores

2 Select host

3 Ready to complete

Select datastores

Select datastores to place the secondary VM disks and configuration files.

Configure per disk ☐

Name	Capacity	Provisioned	Free	Type
 Datastore-shared	59.37 GB	28.99 GB	30.38 GB	NF

Compatibility:

✓ Compatibility checks succeeded.

CANCEL

BACK

NEXT

Figure 1.25: Fault Tolerance Architecture 2

1.14 Project Overview

This section gives an overview of the project, showing its snapshots, cloning process, and template layout. Each part highlights a key aspect of the project.

1.14.1 Snapshots

Here we present the main snapshots of the project. These images illustrate the interface, key steps, and overall functionality.

Manage Snapshots | centos-01

centos-01

VM Snapshot 2%252f6%252f2026, 9:00:00 PM

You are here

Name	VM Snapshot 2%252f6%252f2026, 9:00:00 PM
Created	02/06/2026, 9:00:04 PM
Disk usage	16 GB
Snapshot the virtual machine's memory	No
Quiesce guest file system	No

EDIT

DELETE ALL

DELETE

REVERT TO

DONE

Figure 1.26: Snapshot 1: Description here

1.14.2 Clone

This part explains the cloning process. It shows how to replicate the project environment or setup on another machine.

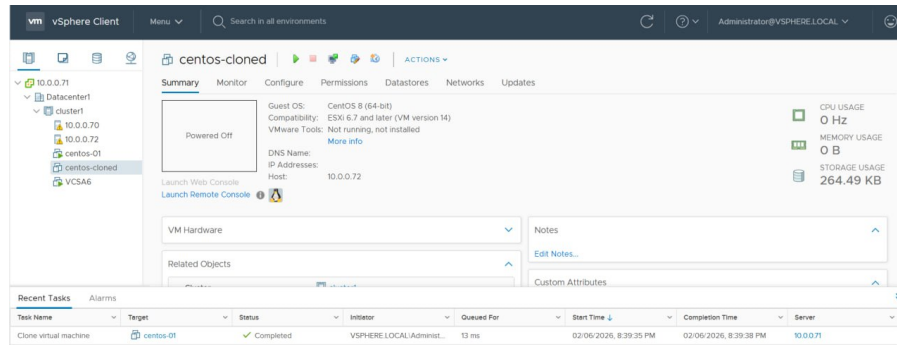


Figure 1.27: Cloning process screenshot

1.15 Abbreviations

Table 1.1: List of Abbreviations

+ Term	Full Form
API	Application Programming Interface
CPU	Central Processing Unit
DHCP	Dynamic Host Configuration Protocol
DNS	Domain Name System
DPM	Distributed Power Management
FDM	Fault Domain Manager
FT	Fault Tolerance
HA	High Availability
iSCSI	Internet Small Computer System Interface
KVM	Kernel-Based Virtual Machine
MAC	Media Access Control
MOID	Managed Object ID
NAS	Network Attached Storage
NFS	Network File System
NIC	Network Interface Card
OS	Operating System
OVA	Open Virtual Appliance
OVF	Open Virtual Format
SSH	Secure Shell
UI	User Interface
VDS	vSphere Distributed Switch
VHD	Virtual Hard Disk
VLAN	Virtual Local Area Network
VMDK	VMware Virtual Disk
VMFS	Virtual Machine File System
VMs	Virtual Machines
vDrs	Virtual Distributed Resource Scheduler
VSAN	Virtual SAN